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ORGANIZATION OF ISLAMIC COOPERATION



SWE 4602

Software Design and Architecture Lab

Lab 03 Report

Team 05:

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Introduction

Money Mate is a **full-stack personal-finance and peer-to-peer loan tracker** built with Next.js, Tailwind CSS, and a MySQL database. The following sections summarize its architectural design, component-level design, user-interface design, and illustrate a typical usage scenario.

1. Architectural Design (High-Level)

Layer	Technologies	Key Responsibilities
Presentation	Next.js (React, Pages/App Router), Tailwind CSS	Responsive UI, server-side rendering (SSR) for SEO, client-side routing, form handling
API & Services	Next.js API routes, JWT guard middleware	RESTful endpoints for auth, transactions, loans, blogs; token verification; rate limiting
Business Logic	Service classes / Domain models (e.g., TransactionService, LoanService)	Validation, calculations (budget, interest, overdue), orchestration of notifications
Data Access	Prisma ORM (MySQL)	Schema definitions, migrations, type-safe queries
Infrastructure & Integrations	SendGrid, Stripe/Razorpay, Google Calendar API (future), S3 (images)	Email notifications, payment handling, optional calendar export, file storage
Cross-Cutting Concerns	NextAuth + JWT, HTTPS, centralized error logging, caching	Security, performance, reliability, monitoring

Communication follows a classic **N-tier architecture**: the browser calls Next.js pages/APIs, which invoke services, which in turn persist data via Prisma to MySQL. External integrations are accessed through dedicated adapters, keeping the core domain clean.

2. Component-Level Design

Component	Major Subcomponents	Data Entities
User Management	- Registration/Login - JWT refresh flow - Profile service	User, SessionToken

Income & Expense Tracker	- Transaction CRUD - Category tagging - Monthly summary generator	Transaction, Category
Loan Management	- Loan CRUD - Installment scheduler - Reminder engine	Loan, Installment, Payment
Budget Planner	- Budget CRUD - Threshold checker - Alert dispatcher	Budget, BudgetCategory
Insights & Analytics	- Chart data builders - Trend calculators	Aggregated views on Transaction / Loan
Notification Center	- In-app alert queue - Email sender (SendGrid) - Socket hub (optional)	Notification
Community Blogs	- Blog CRUD - Like/Comment services - Sorting & search	BlogPost, Comment, Like, Tag

Each component exposes a **service interface** used by API route controllers. Dependency injection keeps components testable and allows mocking in unit tests.

3. User-Interface Design

1. **Dashboard (Home)**
    - At-a-glance cards: current balance, month-to-date income vs. expenses, active loans, budget status.
    - Quick-add buttons for expense, income, or loan.
2. **Transactions**
    - Tabbed view: *Income* and *Expenses*.
    - Filter chips (date range, category) and searchable table.
    - Modal form with smart suggestions for categories and payees.

3. **Loans**
    - Kanban-style columns: Active, Due Soon, Overdue, Paid.
    - Detail drawer shows installment timeline and partial-payment form.
4. **Budget Planner**
    - Radial progress per category with color cues (green, amber, red).
    - “Create Budget” wizard pre-loads average historical spend.
5. **Insights**
    - Interactive bar, line, and pie charts rendered via a lightweight charting lib.
    - Time-range picker syncs across all graphs.
6. **Community Blog**
    - Masonry grid of posts; sort dropdown (Most liked, Commented, Recent).
    - Rich-text editor with markdown preview for authors.
    - Inline like & comment actions, showing counts and avatars.
7. **Navigation & Responsiveness**
    - Left collapsible sidebar on desktop; bottom tab bar on mobile.
    - Dark-mode toggle and dynamic theming using Tailwind CSS variables.

4. Software Usage Scenario

Persona: Ayesha, 28, freelance graphic designer managing irregular income and occasional loans among friends.

1. **Onboarding**
   Ayesha signs up, sets her preferred currency (BDT), and is greeted by a clean dashboard with zero data.

2. **Capturing Income & Expenses**

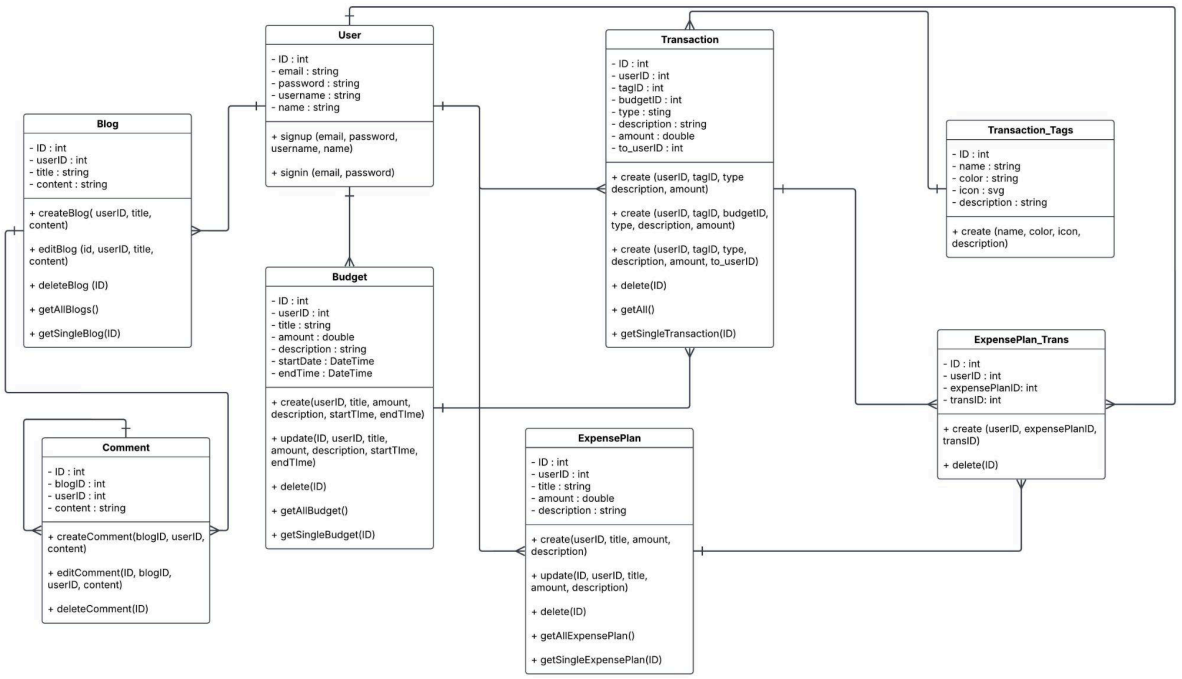
- She records last week's project payment as income (₮35,000, "Client X").
- Throughout the week she logs expenses—food, ride-sharing, subscription fees.

3. **Budget Setup**
    - She creates a monthly budget of ₮25,000 for discretionary spending.
    - Money Mate now tracks her spend and shows a radial gauge reaching 60% halfway through the month.
4. **Peer-to-Peer Loan**
    - A friend borrows ₮5,000; Ayesha enters a *Lent* loan, sets two weekly installments.
    - The friend repays one installment; Ayesha records the partial payment, reducing the outstanding balance.
5. **Reminders & Notifications**
    - Two days before the next installment, Money Mate sends an in-app alert and an email reminder.
    - A budget alert triggers when her entertainment category nears its limit.
6. **Insights Review**
    - At the end of month, Ayesha views charts showing income vs. expenses, and identifies overspending on dining.
    - She adjusts next month's budget accordingly.
7. **Community Engagement**
    - She writes a blog post sharing tips on freelancing cash-flow management.
    - Fellow users like and comment, boosting her engagement stats visible on her profile.

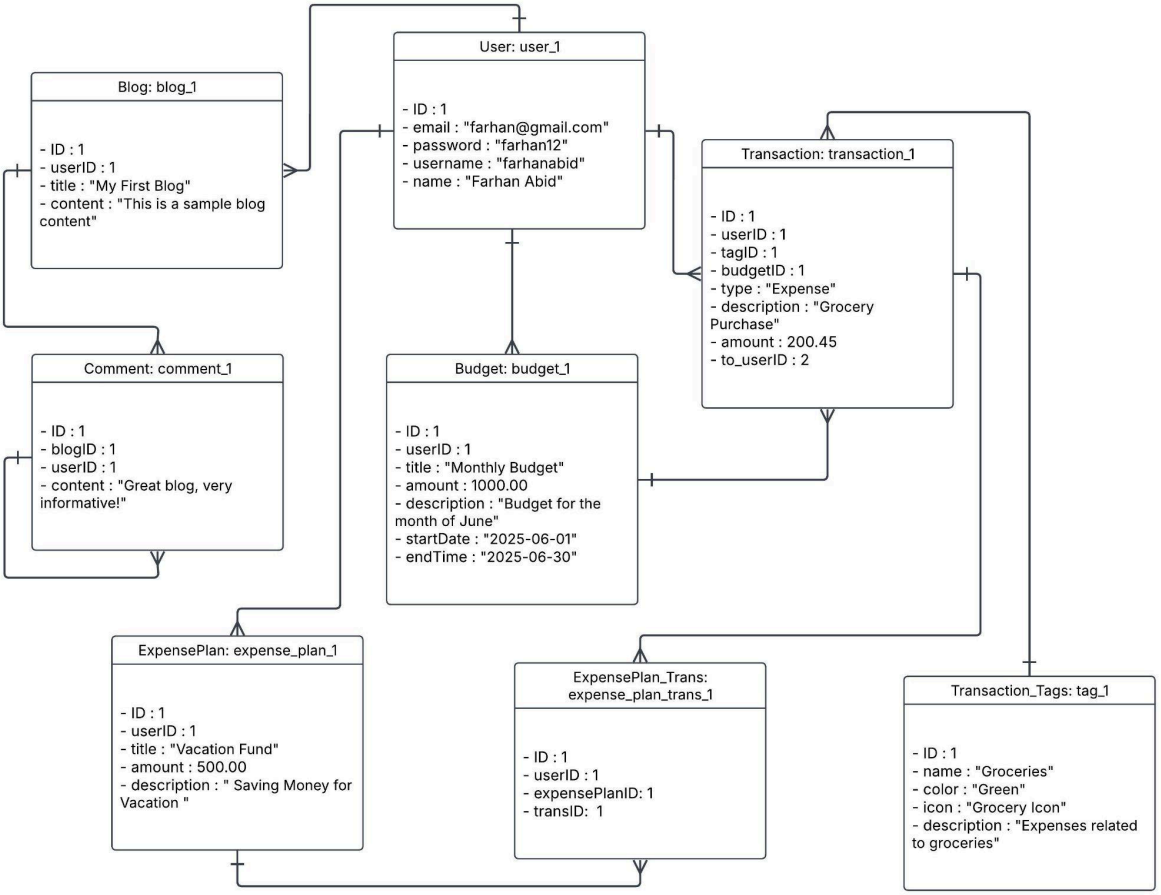
Throughout the cycle, API responses remain under 500 ms, and the app maintains a minimalist, mobile-first experience with clear visual hierarchy.

Structural Design

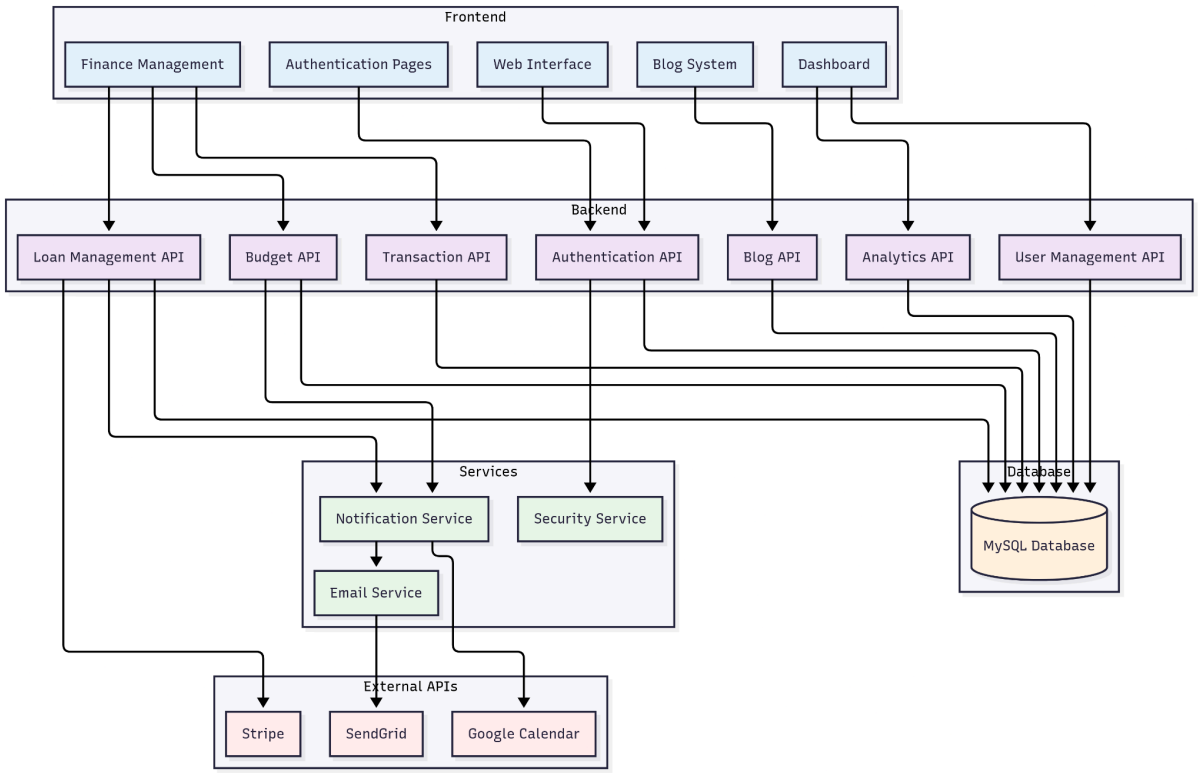
Class Diagram



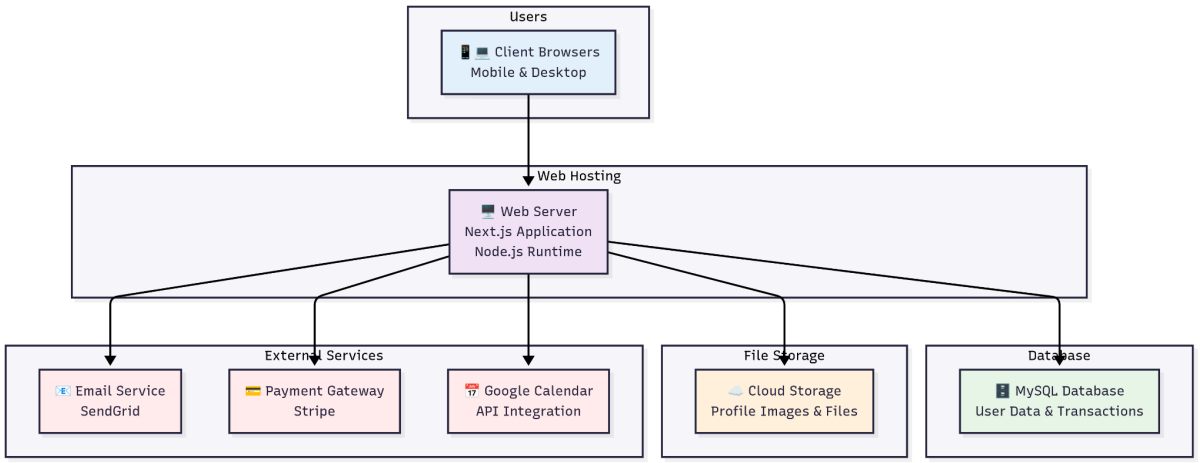
Object Diagram



Component Diagram

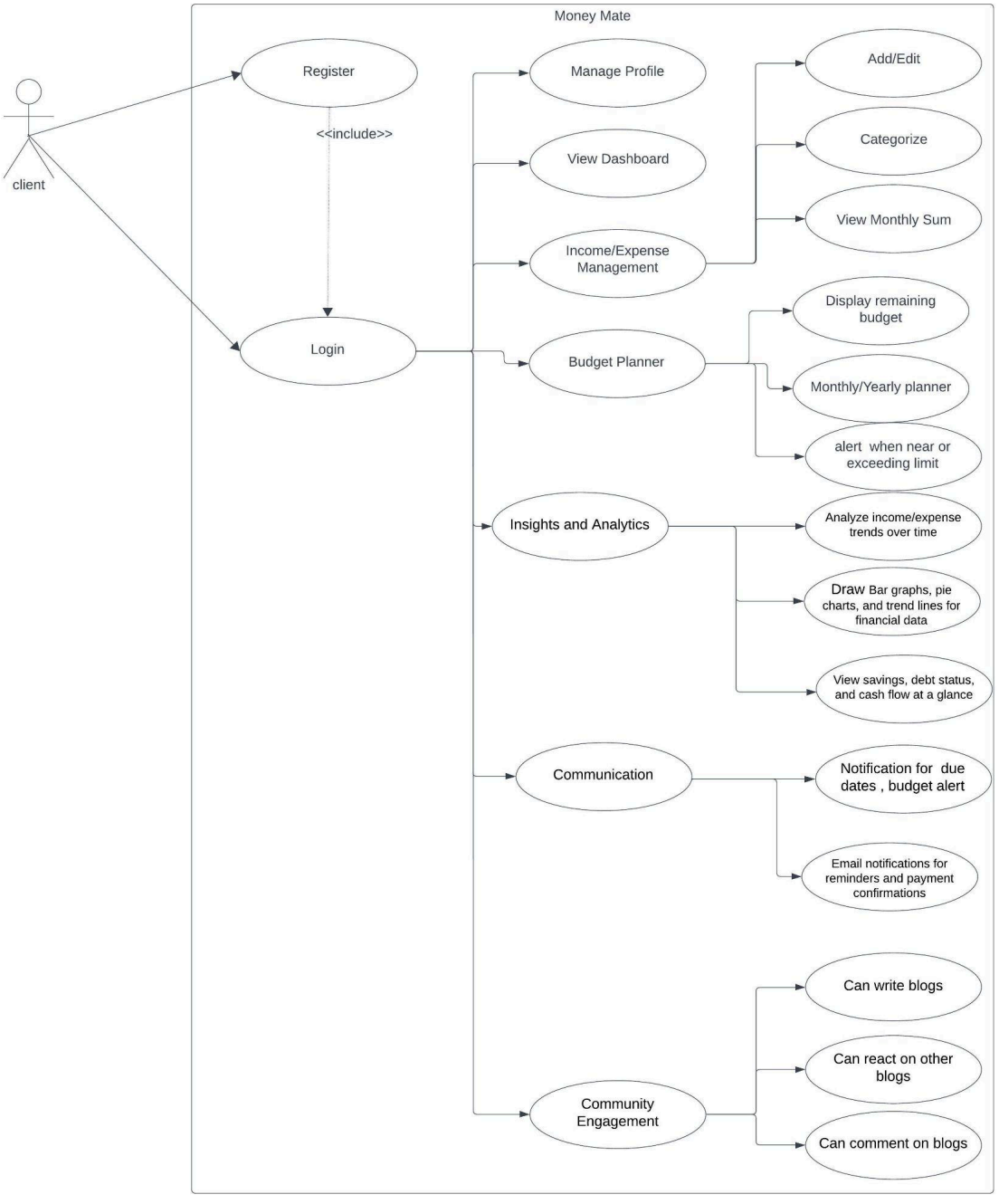


Deployment Diagram

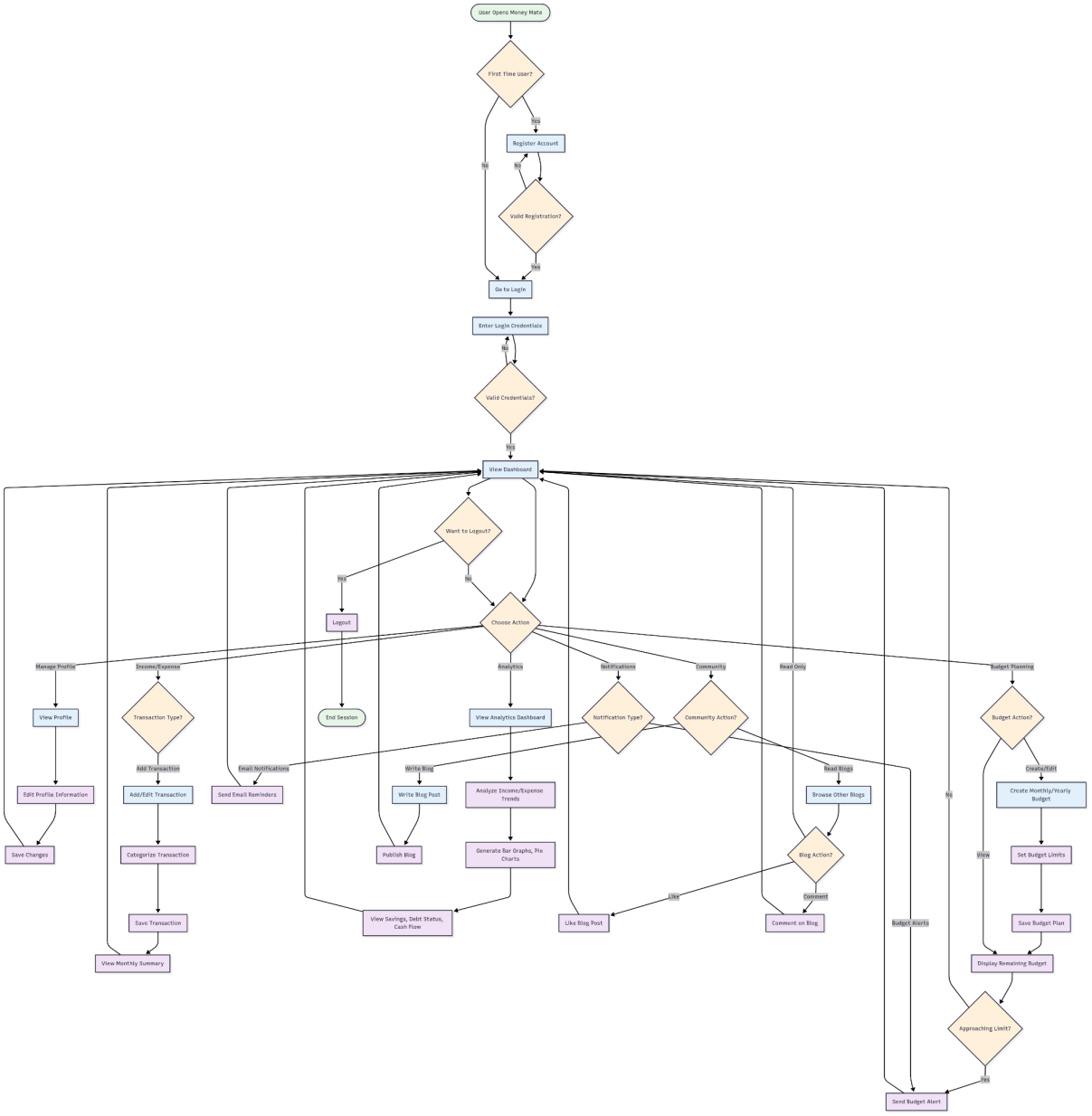


Behavioral Design

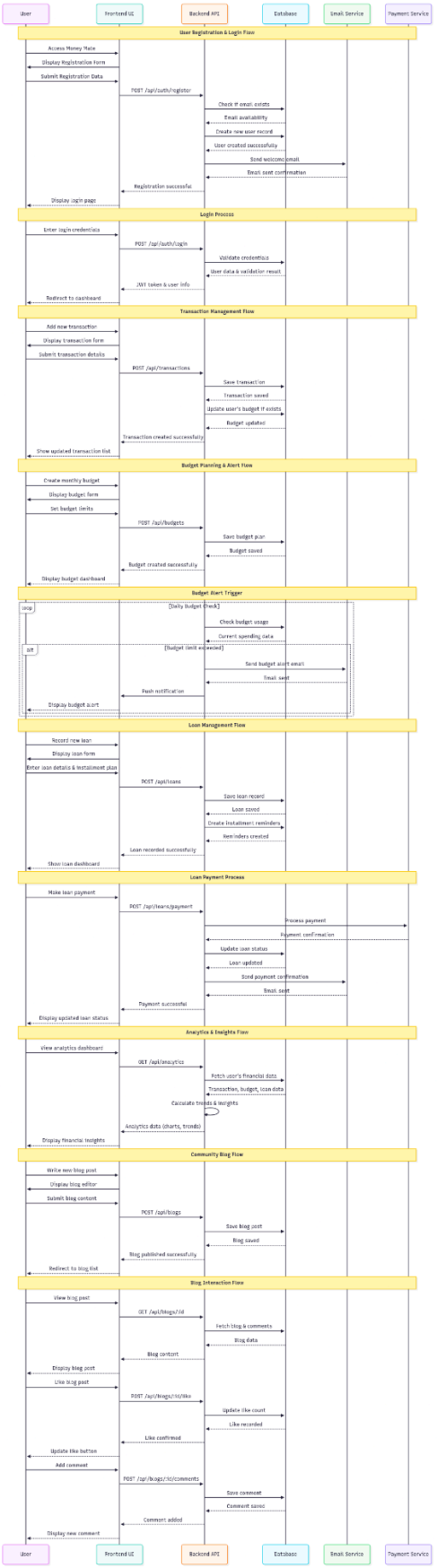
Use Case Diagram



Activity Diagram



Sequence Diagram



User Interface Design - User Analysis

1. Primary User Personas

Persona 1: The Young Professional (Sarah, 26)

Demographics:

- Age: 24-30
- Occupation: Software developer, marketing coordinator, junior analyst
- Income: $35,000-$60,000 annually
- Tech Savviness: High
- Location: Urban areas

Goals & Motivations:

- Track expenses to save for major purchases (car, house down payment)
- Build healthy financial habits early in career
- Manage money lent to friends and roommates
- Understand where money is being spent

Pain Points:

- Forgets to track small expenses
- Difficulty managing shared expenses with friends
- Lacks financial literacy for long-term planning
- Uses multiple apps that don't integrate well

Behavioral Patterns:

- Primarily mobile-first user
- Values visual data representation
- Appreciates automation and smart categorization
- Active on social platforms, values community features

Persona 2: The Budget-Conscious Family Manager (Michael, 34)

Demographics:

- Age: 30-45
- Occupation: Teacher, retail manager, administrative roles
- Income: $40,000-$70,000 annually
- Tech Savviness: Medium
- Family Status: Married with 1-2 children

Goals & Motivations:

- Control household expenses and stick to budget
- Plan for children's education and family emergencies
- Track loans from family members or friends
- Achieve specific savings goals (vacation, home improvement)

Pain Points:

- Multiple income sources and complex expense categories
- Difficulty involving spouse in financial tracking
- Needs simple, non-technical interface
- Wants quick overview without detailed analysis

Behavioral Patterns:

- Uses both mobile and desktop
- Prefers straightforward, functional design
- Values reliability over advanced features
- Needs clear alerts and reminders

Persona 3: The Financially Conscious Student (Emma, 21)

Demographics:

- Age: 18-25
- Occupation: College student with part-time job
- Income: $12,000-$25,000 annually (including financial aid)
- Tech Savviness: High
- Location: College towns

Goals & Motivations:

- Manage limited budget effectively
- Track money borrowed from/lent to friends
- Build credit and financial responsibility
- Learn financial management skills

Pain Points:

- Very limited income with irregular patterns
- Frequent small peer-to-peer transactions
- Needs free or very low-cost solutions
- Limited financial knowledge

Behavioral Patterns:

- Mobile-exclusive user
- Values peer learning and community features
- Responsive to gamification elements
- Price-sensitive

2. Secondary User Groups

Small Business Owners

- Need expense tracking for business purposes
- Require loan management for business relationships
- Value analytics for business planning

Retirees on Fixed Income

- Focus on expense monitoring
- Less tech-savvy, need simple interfaces
- Value security and reliability

Freelancers/Gig Workers

- Irregular income patterns
- Need detailed expense categorization
- Require analytics for business planning

3. User Needs Analysis

Core Functional Needs

1. Expense Tracking
    - Quick entry of transactions
    - Automatic categorization
    - Receipt attachment capability
2. Income Management
    - Multiple income source tracking
    - Irregular income handling
    - Projection capabilities
3. Peer-to-Peer Loan Management
    - Easy loan creation and tracking
    - Payment reminders
    - Dispute resolution features

4. Budgeting
    - Flexible budget creation
    - Real-time tracking
    - Alert systems
5. Analytics & Insights
    - Visual data representation
    - Trend analysis
    - Goal tracking

Emotional Needs

- **Security**: Trust in data protection and financial privacy
- **Control**: Feeling empowered about financial decisions
- **Clarity**: Understanding financial situation at a glance
- **Achievement**: Progress toward financial goals
- **Peace of Mind**: Confidence in financial tracking accuracy

Social Needs

- **Accountability**: Community support for financial goals
- **Learning**: Access to financial education and tips
- **Sharing**: Ability to discuss strategies (anonymously)
- **Recognition**: Achievement acknowledgment

4. User Journey Mapping

New User Journey

1. **Discovery** → App store search, friend recommendation, social media
2. **Registration** → Simple signup process, profile setup
3. **Onboarding** → Tutorial, initial data entry, goal setting
4. **First Use** → Adding first transactions, exploring features
5. **Regular Use** → Daily/weekly transaction entry, budget monitoring
6. **Advanced Use** → Analytics exploration, community engagement

Returning User Journey

1. **Quick Entry** → Fast transaction logging
2. **Status Check** → Dashboard review, budget status
3. **Analysis** → Monthly/weekly insights review
4. **Goal Adjustment** → Budget modifications, goal updates

5. **Community Engagement** → Blog reading, commenting

5. Device & Platform Preferences

Mobile Usage (Primary - 80%)

- Quick transaction entry during/after purchases
- Photo capture for receipts
- Push notifications for reminders
- Location-based expense categorization

Desktop Usage (Secondary - 20%)

- Detailed analytics review
- Budget planning sessions
- Loan management setup
- Community blog writing

6. Accessibility Requirements

Technical Accessibility

- Screen reader compatibility
- High contrast mode support
- Large font options
- Voice input capabilities

Cognitive Accessibility

- Simple navigation structure
- Clear visual hierarchy
- Consistent terminology
- Progressive disclosure of features

Physical Accessibility

- Touch-friendly interface elements
- Gesture alternatives
- Voice control options

7. User Success Metrics

Engagement Metrics

- Daily/Weekly active users
- Session duration
- Feature adoption rates
- Retention rates (30, 60, 90 days)

Financial Health Metrics

- Budget adherence rates
- Savings goal achievement
- Loan repayment tracking accuracy
- Expense categorization accuracy

Community Metrics

- Blog post creation and engagement
- User-generated content quality
- Peer interaction frequency
- Knowledge sharing effectiveness

8. User Support Requirements

Self-Service Support

- Comprehensive help documentation
- Video tutorials
- FAQ sections
- Community forums

Direct Support

- Chat support for technical issues
- Email support for account problems
- Phone support for security concerns

Educational Content

- Financial literacy articles
- Best practices guides
- Goal-setting workshops

- Budgeting templates

User Interface Design - Task Analysis

1. Task Hierarchy

Primary Tasks (Core Functionality)

1. Financial Transaction Management
2. Budget Planning and Monitoring
3. Peer-to-Peer Loan Management
4. Financial Analytics and Insights
5. Account and Profile Management

Secondary Tasks (Supporting Features)

6. Community Engagement
7. Notification Management
8. Data Export and Backup

2. Detailed Task Analysis

Task 1: Financial Transaction Management

1.1 Add Income Transaction

Goal: Record money received from various sources **Frequency:** Weekly to monthly **User Types:** All users **Priority:** High

Pre-conditions:

- User is logged in
- User has received income

Main Flow:

1. Navigate to income section or dashboard
2. Click "Add Income" button
3. Enter amount
4. Select/enter income source
5. Set date (default: today)
6. Add description/notes (optional)

7. Attach receipt/proof (optional)
8. Save transaction

Post-conditions:

- Income recorded in database
- Dashboard updated with new total
- Budget calculations refreshed

Alternative Flows:

- Quick add from dashboard widget
- Voice input for amount and source
- Recurring income setup

Error Conditions:

- Invalid amount format
- Missing required fields
- Network connectivity issues

1.2 Add Expense Transaction

Goal: Record money spent on various items/services **Frequency:** Daily **User Types:** All users **Priority:** High

Pre-conditions:

- User is logged in
- User has made a purchase

Main Flow:

1. Navigate to expense section or use quick-add
2. Click "Add Expense" button
3. Enter amount
4. Select category from dropdown or create new
5. Set date (default: today)
6. Add description
7. Take photo of receipt (optional)
8. Add tags (optional)
9. Save transaction

Post-conditions:

- Expense recorded and categorized
- Budget updated with remaining amounts

- Analytics data refreshed

Alternative Flows:

- Camera-first entry (photo then details)
- Location-based auto-categorization
- SMS/email import

Error Conditions:

- Camera permission denied
- Category selection fails
- Duplicate transaction warning

1.3 Edit Transaction

Goal: Modify existing transaction details **Frequency:** Weekly **User Types:** All users **Priority:** Medium

Main Flow:

1. Find transaction in list/dashboard
2. Click/tap transaction to open details
3. Click "Edit" button
4. Modify desired fields
5. Save changes
6. Confirm changes

Business Rules:

- Users can only edit their own transactions
- Transactions older than 30 days may require admin approval
- Editing affects budget calculations retroactively

Task 2: Budget Planning and Monitoring

2.1 Create New Budget

Goal: Set spending limits for different categories **Frequency:** Monthly or yearly **User Types:** All users **Priority:** High

Pre-conditions:

- User is logged in
- User understands their income patterns

Main Flow:

1. Navigate to Budget section
2. Click "Create New Budget"
3. Select budget period (monthly/yearly)
4. Set total budget amount
5. Allocate amounts to expense categories
6. Set alert thresholds (e.g., 80% of limit)
7. Review and save budget

Post-conditions:

- Budget plan created and activated
- Dashboard shows budget vs. actual spending
- Alert system activated

Validation Rules:

- Total category allocations ≤ total budget
- All amounts must be positive numbers
- At least one category must be defined

2.2 Monitor Budget Status

Goal: Check current spending against budget limits **Frequency:** Daily to weekly **User Types:** All users **Priority:** High

Main Flow:

1. View dashboard or navigate to budget section
2. Review spending by category
3. Check remaining amounts
4. Identify over-budget categories
5. View trend analysis

Information Displayed:

- Visual progress bars for each category
- Remaining amount/days in budget period
- Spending trends and patterns
- Alerts for approaching limits

Task 3: Peer-to-Peer Loan Management

3.1 Record New Loan

Goal: Document money lent to or borrowed from someone **Frequency:** Monthly **User Types:** All users **Priority:** Medium

Pre-conditions:

- User is logged in
- Loan agreement exists (formal or informal)

Main Flow:

1. Navigate to Loans section
2. Click "Add New Loan"
3. Select loan type (lent/borrowed)
4. Enter counterparty information
5. Enter loan amount
6. Set repayment terms
7. Add loan purpose/description
8. Set up payment schedule (optional)
9. Save loan record

Post-conditions:

- Loan tracked in system
- Reminders set up if applicable
- Dashboard updated with loan status

3.2 Record Loan Payment

Goal: Update loan with partial or full payment

Frequency: As payments occur

User Types: All users

Priority: Medium

Main Flow:

1. Find loan in active loans list
2. Click "Record Payment"
3. Enter payment amount

4. Set payment date
5. Add payment method (optional)
6. Save payment record
7. View updated loan balance

Post-conditions:

- Loan balance updated
- Payment history recorded
- Notifications sent if loan is fully paid

3. Task Complexity Analysis

High Complexity Tasks

- **Budget Creation with Multiple Categories:** Requires financial planning knowledge
- **Loan Management with Installments:** Complex calculations and scheduling
- **Advanced Analytics:** Data interpretation skills needed

Medium Complexity Tasks

- **Transaction Entry with Categorization:** Requires consistency in categorization
- **Report Generation:** Understanding of financial data presentation

Low Complexity Tasks

- **Basic Transaction Entry:** Simple data input
- **Profile Updates:** Standard form completion
- **Community Engagement:** Familiar social media patterns

4. Task Frequency and Priority Matrix

Task	Frequency	Priority	Complexity
Add Expense	Daily	High	Low
Add Income	Weekly	High	Low
Check Budget Status	Daily	High	Low
Create Budget	Monthly	High	Medium
View Analytics	Weekly	Medium	Medium

Manage Loans	Monthly	Medium	High
Community Engagement	Weekly	Low	Low
Generate Reports	Monthly	Low	Medium

5. User Interface Implications

High-Priority Quick Access

- Expense entry (floating action button)
- Budget status (dashboard widget)
- Transaction list (main navigation)

Secondary Access

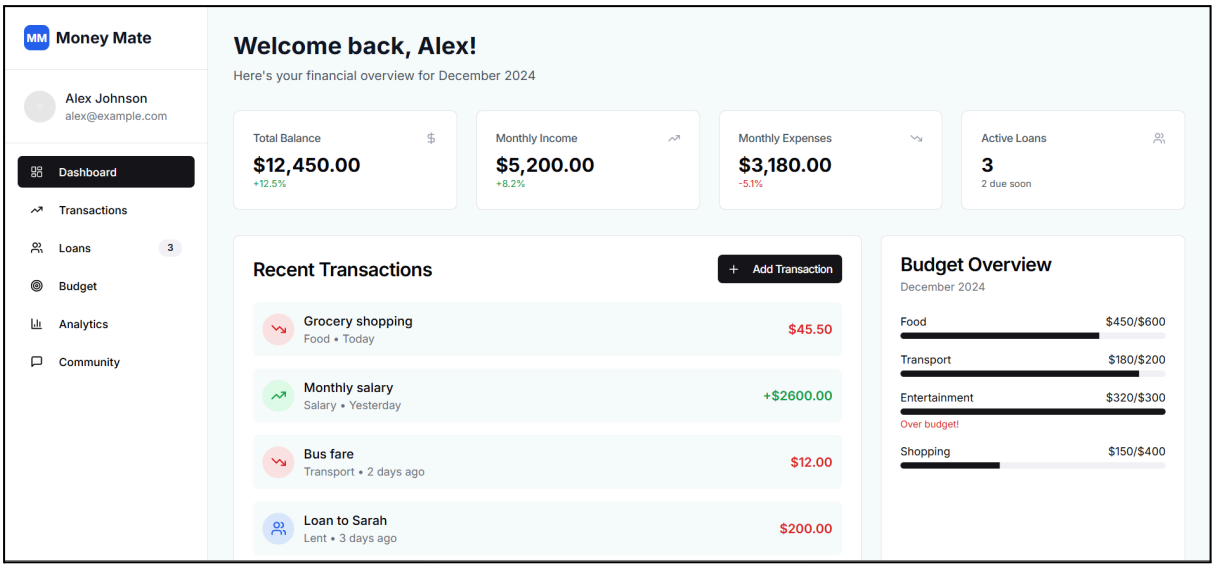
- Analytics (main navigation)
- Loan management (main navigation)
- Community features (main navigation)

Settings/Profile Access

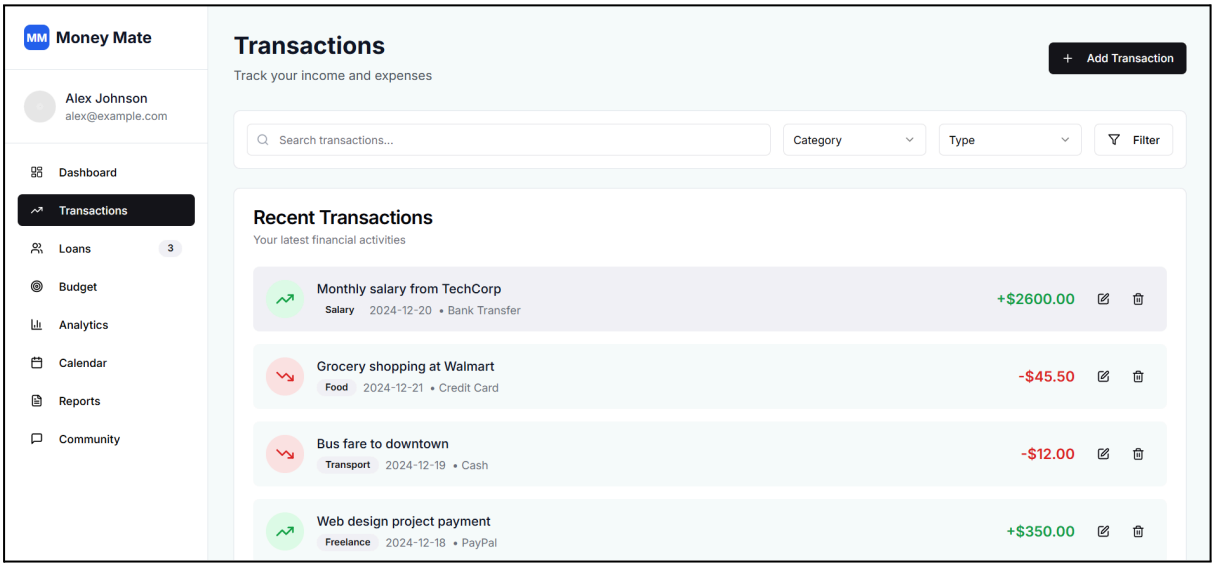
- Profile management (settings menu)
- Preferences (settings menu)
- Help and support (settings menu)

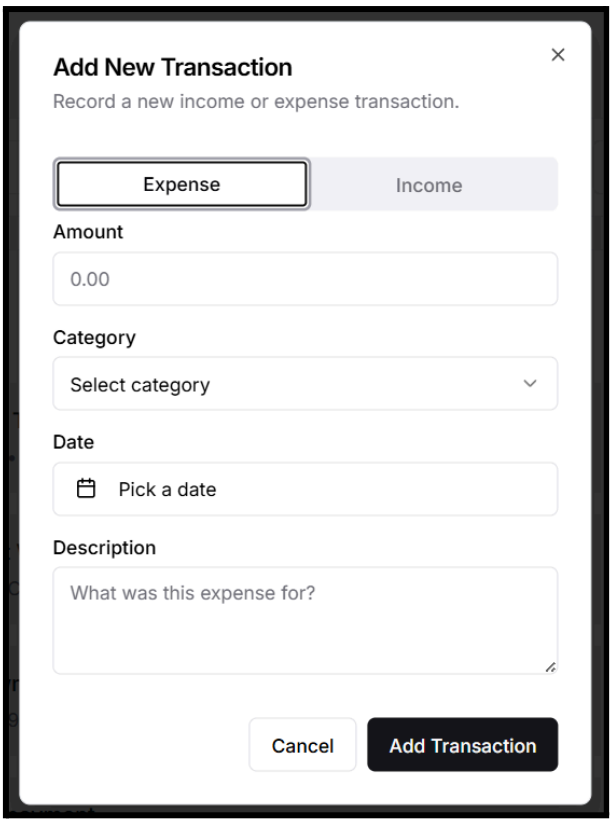
Demo Interfaces

Dashboard

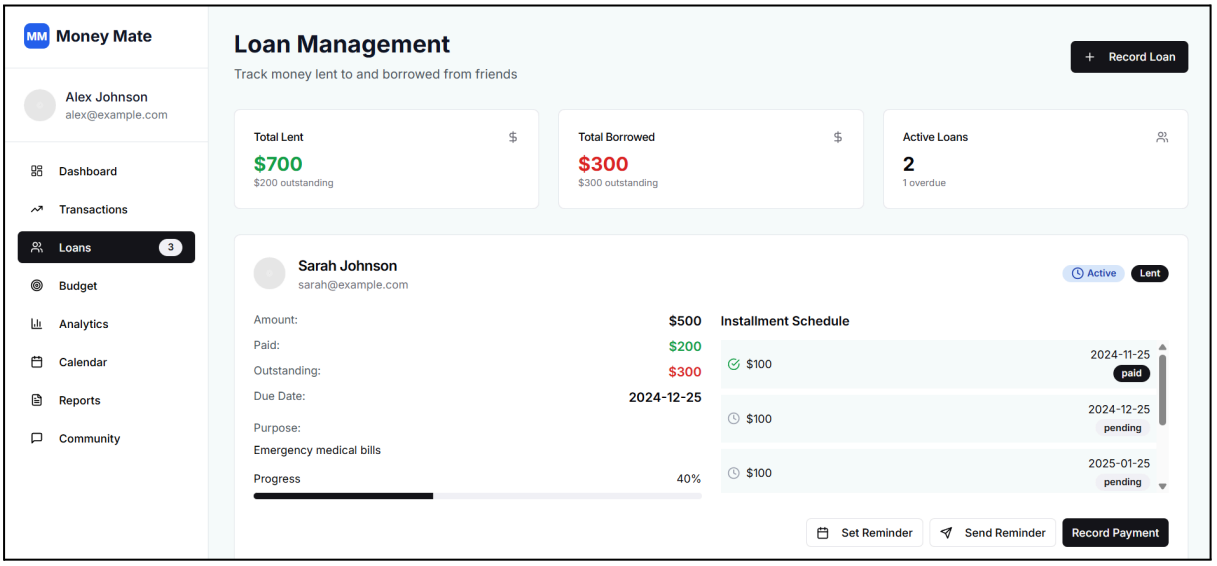


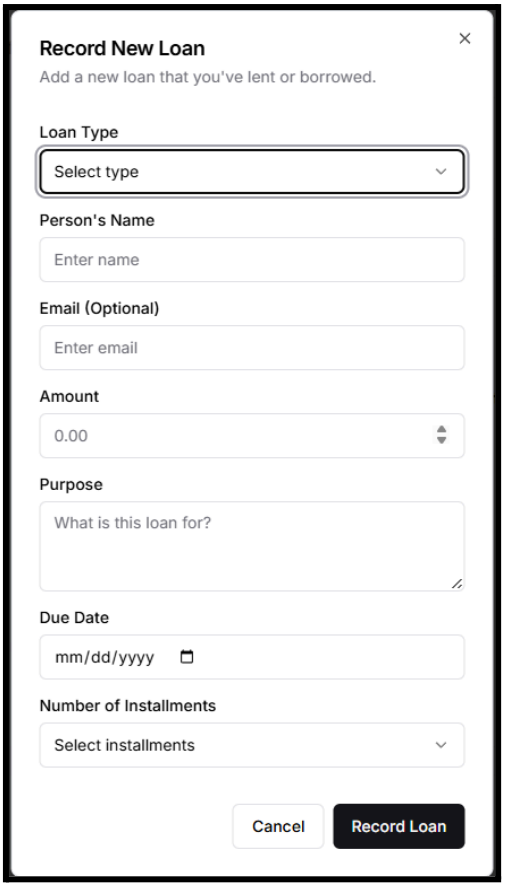
Transaction Management



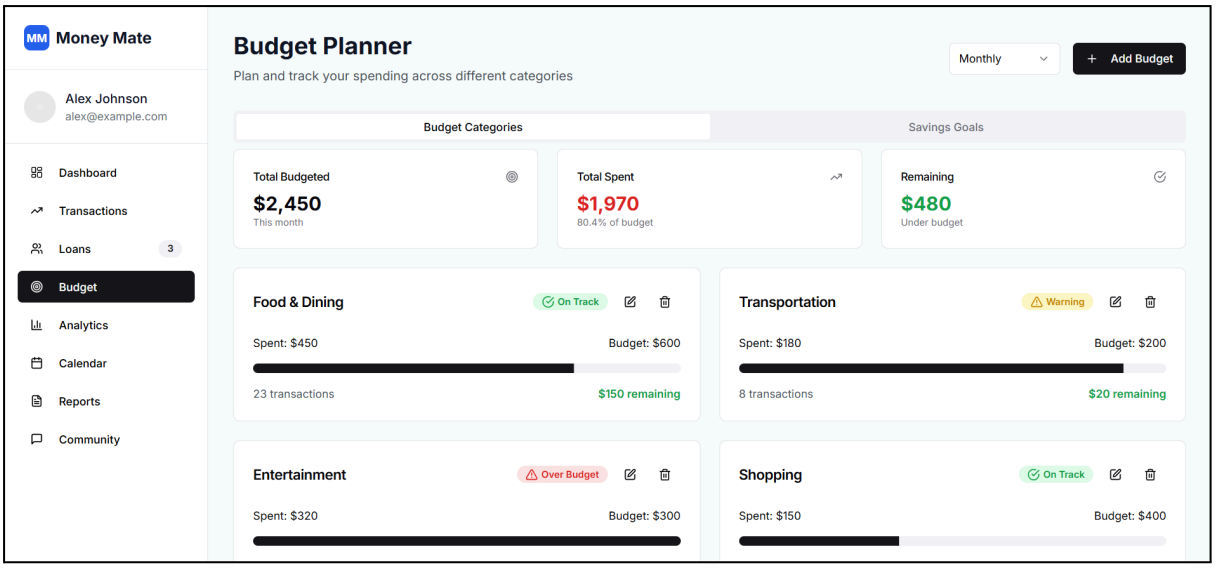


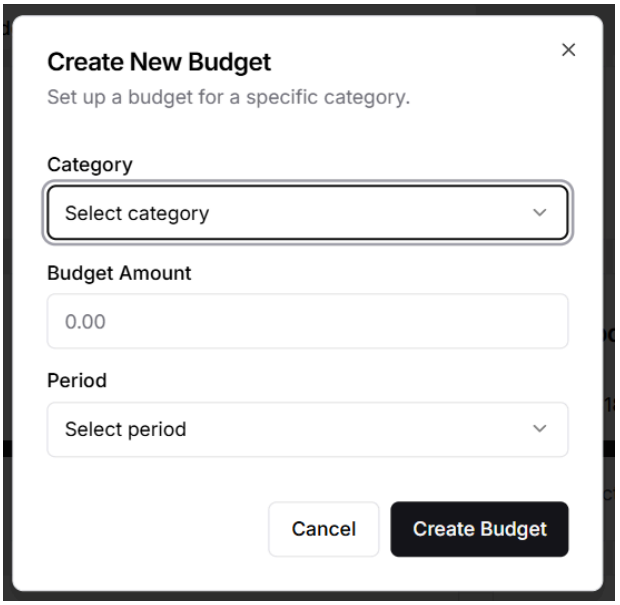
Loan Management



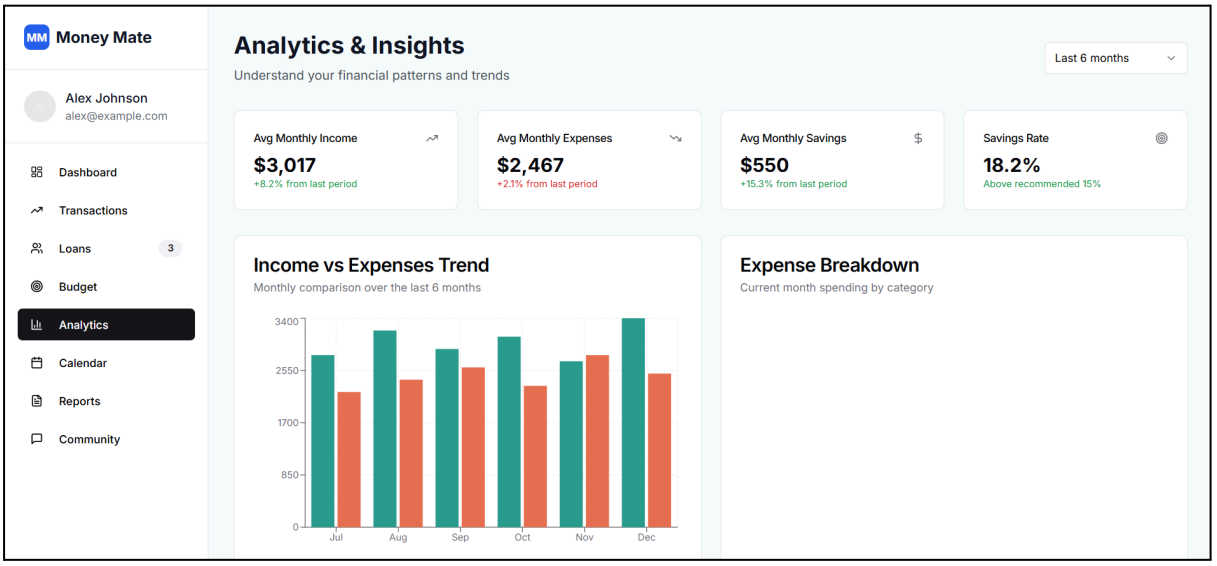


Budget Planner

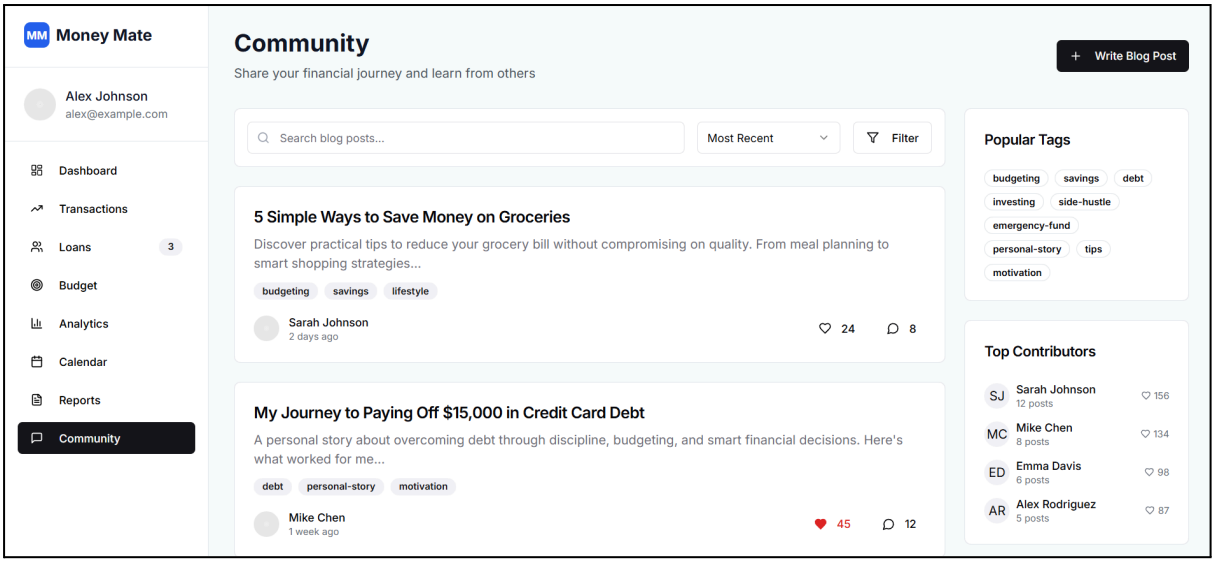




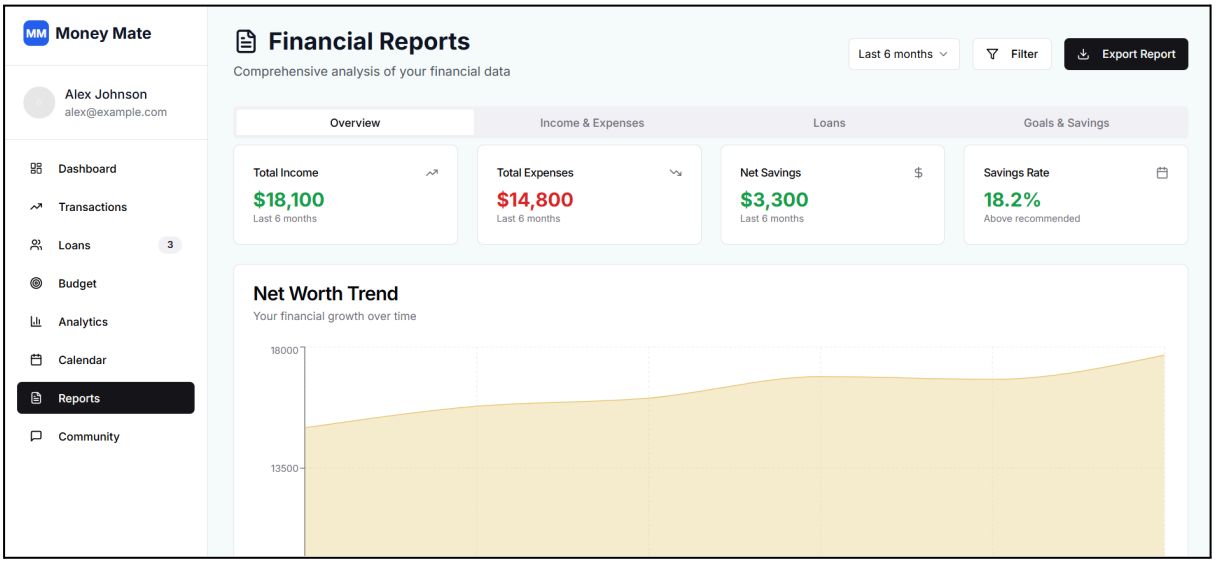
Analytics and Insights



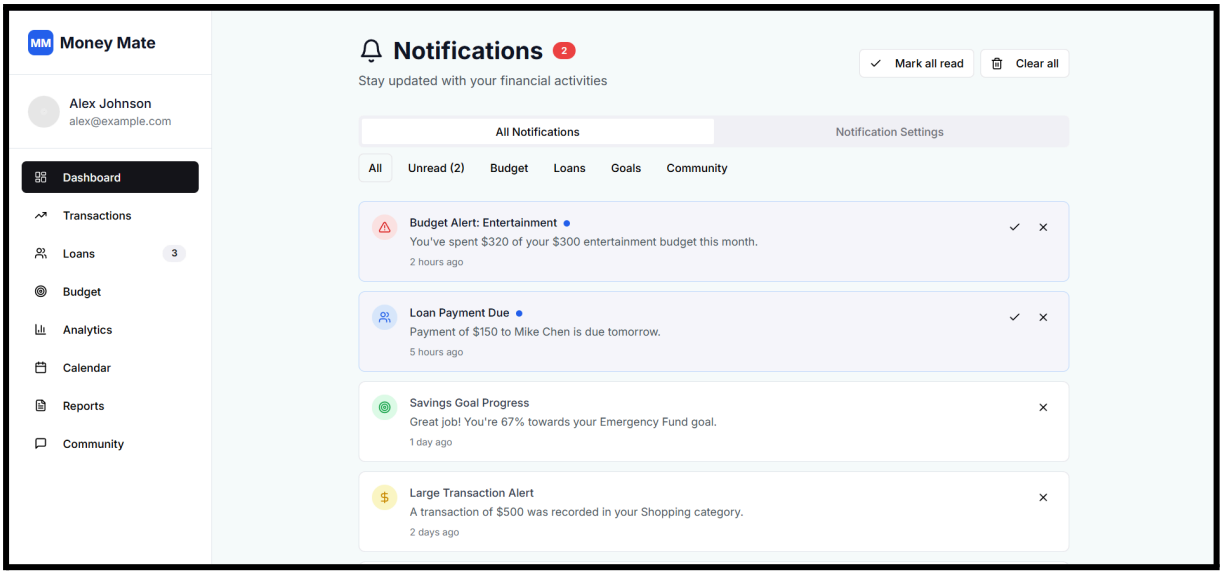
Community Engagement



Reports



Notification



Settings

